What Can Be Guaranteed When an Adaptive System Decides How Much of a Blood Smear to Examine?

Information-gain stopping, rare-event lower bounds, and certified blast-threshold decisions

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Abstract

Automated examination of peripheral blood smears increasingly selects which fields to inspect and when to stop. We ask what such a procedure can guarantee about the part of the specimen it did *not* examine. Using a pre-registered protocol (four frozen amendments; about 66,000 confirmatory simulated specimens, most of them built from 28,280 expert-labelled cells; an external dataset; and real, fully annotated smears), we obtain four results. **(i)** A reimplementation of information-saturation stopping (InfoDPP-PAC) from its published equations halts at the same point, about 16% of the specimen, regardless of how many decision-changing fields are present. Its halting signal loses its association with blasts as cells are pooled into fields (AUC 0.78 at one cell per field, 0.52 at twenty), while the relevance score it ignores remains informative (0.98 to 0.90). With one critical field among 200 and ten cells per field, it misses 76% of positive specimens. **(ii)** We show that this is not specific to the method: when pre-inspection features carry no information about where rare evidence lies, every strategy, adaptive or not, has worst-case miss probability at least $\binom{N-k}{n}/\binom{N}{n}$. Certifying the absence of a single critical field at $\alpha = 0.05$ therefore requires examining 95% of the fields. **(iii)** Reformulating the question as a bounded-mean assertion ("the blast proportion exceeds the diagnostic threshold") makes certification feasible. Using ALPHA and importance-weighted betting tests, false certification stayed between 0.001 and 0.022 at thresholds of 0.10 (ICC 2022) and 0.20 (WHO 2022). Classifier-driven sampling reduced the number of cells examined by 25–55%, whereas an uncorrected adaptive design certified falsely in 89–99% of boundary cases. A conventional 100-cell count called "above" in 41–43% of specimens lying exactly on the threshold. **(iv)** Validity survived a change of laboratory that reduced classifier blast recall from 0.91 to 0.34, but efficiency did not: classifier-driven sampling then needed more cells than uniform sampling (195 vs. 110). In real bone-marrow smears, blasts are measurably clustered within fields. Analysing the cells of sampled fields as individual draws then inflates false certification, modestly at the measured clustering (model-based: about 0.05–0.10) and severely under strong clustering (0.33–0.48). In short, guarantees about unexamined material come from the sampling design and the unit of analysis, not from model confidence. We state explicitly what our experiments do not establish, including clinical validity.

1 Introduction

A peripheral blood smear contains far more cells than any examiner inspects. A manual differential usually counts 100 or 200 leukocytes, and the statistical variability that this implies has been known for decades [1]. Automated systems add a new degree of freedom: they choose which fields to look at and decide when enough has been seen. For diagnostic questions that hinge on rare cells, such as the proportion of blasts relative to a classification threshold [2, 3], the relevant question is not how accurate the classifier is on the fields it examined, but what can be said about the fields it did not examine.

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Adaptive stopping rules based on *information saturation* address this question implicitly. They stop when additional fields are expected to add little information about the specimen as the selector's embedding represents it. InfoDPP-PAC [4] is a recent, carefully constructed instance for whole-slide histopathology. It combines Gaussian-process relevance, determinantal diversity and a PAC-style halting certificate, and its authors state clearly that the certificate bounds residual information in the selection objective, not downstream diagnostic performance. That distinction is the starting point of this paper. Saturation of information about the embedding does not imply that the decision-relevant evidence has been seen, especially when that evidence is rare.

We study this gap in the setting of peripheral blood smears and blast counting, and ask four research questions:

RQ1 Does information-saturation stopping fail systematically when decision-changing evidence is rare?

RQ2 Why does certifying the *absence* of rare evidence require near-exhaustive examination in the worst case?

RQ3 How many cells must be examined to certify a threshold decision under adaptive sampling?

RQ4 Where do the assumptions of anytime-valid audit methods break on real blood-smear data?

Contributions.

1. A pre-registered evaluation of information-saturation stopping on specimens assembled from real expert-labelled cells with controlled rarity and dilution. It shows that the stopping point does not depend on how much critical evidence is present and that its halting signal decouples from critical evidence as cells are pooled into fields (RQ1). One of the Part I predictions failed (three of the 20 pre-registered hypotheses overall), and these failures are reported as such.

2. A worst-case lower bound, a direct application of Yao's minimax principle [5]. It shows that near-exhaustive examination is required of *every* strategy that lacks location information, not only of uniform sampling (RQ2).

3. An evaluation of design-based, anytime-valid certification of blast-proportion assertions using ALPHA [6] and importance-weighted betting under adaptive sampling [7]. We quantify validity, cost, and the failure of uncorrected adaptive designs and conventional counts (RQ3).

4. An empirical examination of the assumptions behind these methods (RQ4): classifier-error heterogeneity, acquisition shift, and spatial clustering of blasts, the last measured in real smear regions in which every cell was annotated. We separate the assumptions that matter for validity from those that matter only for efficiency.

Scope. This is a methodological study. The "expert" who verifies sampled cells is simulated with the datasets' expert labels, most specimens are composites of real single-cell images, and no result here establishes clinical diagnostic validity.

2 Related work

Variability of cell counts. The binomial and finite-population variability of manual differentials, and the resulting uncertainty of low-frequency cell types, were characterised by Rümke [1]. Our conventional-count baseline makes that variability concrete near a diagnostic threshold.

Adaptive selection with stopping. InfoDPP-PAC [4] selects whole-slide image patches greedily under a blended information-gain and relevance objective, and stops when the maximal information gain among unselected patches falls below a concentration-corrected threshold. It was evaluated on gastrointestinal whole-slide images with selection-quality metrics. We reimplement its halting rule as the object of study, not as a straw man. Automated blast recognition itself is well studied [8]; our question concerns examination and inference, not classification accuracy.

Risk-limiting audits and betting. Risk-limiting audits test assertions about finite populations from a sample, with a guaranteed bound on the probability of confirming a false assertion [9]. ALPHA [6] turns such assertions into bounded-mean tests using an *assorter*, and constructs a test supermartingale for sampling without replacement. Betting-based confidence sequences [10] provide the underlying anytime-valid machinery, through Ville's inequality [11]. Shekhar et al. [7] extend this to weighted, adaptive sampling without replacement: they use importance-weighted observations and allow side information both in the sampling distribution and as control variates. Stratified extensions [12] are relevant when a specimen is naturally partitioned. Prediction-powered inference [13] and prediction-powered e-values [14] combine model predictions with a verified subset. We did not implement the latter; side information enters our tests only through the sampling distribution and control variates.

3 Setting

A specimen consists of N fields, each holding m cells, so there are $M = Nm$ cells in total. Cell j has a blast indicator $b_j \in \{0, 1\}$, and the specimen's blast proportion is $p = M^{-1} \sum_j b_j$. We distinguish three estimands, which the literature sometimes conflates:

1. *Detection*: has at least one critical field (a field with at least one blast) been examined?
2. *Estimation*: what is p?
3. *Threshold certification*: is $p > \theta$, for a stated threshold θ?

We use $\theta = 0.10$, the blast percentage at which the ICC 2022 separates myelodysplastic syndrome with excess blasts from MDS/AML (and, in the presence of AML-defining genetic abnormalities, from AML) [3], and $\theta = 0.20$, the WHO 5th-edition blast threshold for acute myeloid leukemia without defining genetic abnormalities [2]. Both classifications count certain blast equivalents (for example promonocytes); our labels count blast classes only. They are analysed separately and never pooled. These thresholds are inputs to a statistical procedure; the study does not validate them or any diagnosis.

An examiner selects fields sequentially. Selection may use pre-inspection features x (for instance embeddings) and past verified labels, and may be randomised. A selected field is *verified*: the true labels of all its cells are revealed. All probabilities below are over the examiner's randomisation, conditional on the fixed specimen. The guarantees are therefore finite-population and design-based. They hold for every specimen in the stated class and say nothing about a population of patients.

4 Why saturation cannot certify absence (RQ2)

Let $C \subseteq [N]$ be the set of critical fields, with $k = |C|$, and let M_n be the event that none of them is among the (at most) n examined fields. Define

$$h(N, k, n) = \binom{N-k}{n} \Big/ \binom{N}{n} = \prod_{i=0}^{k-1} \frac{N-n-i}{N-i}.$$

Proposition 1 (Uniform sampling)**.** *Under simple random sampling of n fields without replacement, $\mathbb{P}(M_n \mid C) = h(N,k,n)$ exactly. For $k=1$ this equals $1-n/N$, so $\mathbb{P}(M_n) \le \alpha$ if and only if $n \ge \lceil (1-\alpha)N \rceil$. Moreover, $\left(1 - \frac{n}{N-k+1}\right)^k \le h \le (1-n/N)^k$, so $n \ge N(1-\alpha^{1/k})$ suffices.*

Assumption 1 (No location information)**.** For the given features x, every placement C with $|C| = k$ is possible.

Theorem 1 (Minimax lower bound)**.** *Under Assumption 1, every policy (adaptive or not, randomised or not) that examines at most n fields satisfies*

$$\max_{|C|=k} \mathbb{P}_\pi(M_n \mid x, C) \;\ge\; h(N,k,n).$$

Uniform sampling attains this bound for every C. Every deterministic policy has worst-case miss probability 1 whenever $n \le N-k$.

Proof sketch. Fix the examiner's randomness u. On the run in which every examined field is negative, the policy's inputs are determined by u, so it examines a fixed set $A(u)$ with $|A(u)| \le n$. For any C, the actual run coincides with this all-negative run up to the first critical field it would reach, so the examiner misses C exactly when $C \cap A(u) = \emptyset$. Averaging over a uniformly random C, which Assumption 1 permits, gives a miss probability of at least $\binom{N-n}{k}/\binom{N}{k} = h(N,k,n)$. Since the maximum over C is at least this average, the bound follows. The full proof is in Appendix A. □

Corollary 1 (Unknown k)**.** *If an examiner declares "no critical evidence" whenever it has examined none, and this declaration must be wrong with probability at most α for every non-empty C, then, under Assumption 1, it must examine at least $\lceil (1-\alpha)N \rceil$ fields. The reason is that $C \subseteq C'$ implies $M_n(C') \subseteq M_n(C)$ pathwise, so the worst case is $k=1$.*

Remark 1 (Imperfect verification). If verification detects an examined critical field only with probability $1-\beta$, independently of selection, then for $k=1$ under uniform sampling the miss probability is $1-(1-\beta)n/N$, which equals β at $n=N$. Absence therefore cannot be certified at level $\alpha < \beta$ even by a census.

Figure 1 shows the resulting requirement. Informative features (the failure of Assumption 1) allow a search to beat this bound, but only through a model of where critical evidence appears, and that model is not itself certified. Section 7.1 shows how quickly such information disappears when cells are pooled into fields. Section 7.3 shows it failing under acquisition shift. The bound is also specific to detecting the *presence* of rare evidence: making "$p > \theta$" true requires more than θM blasts, which cannot be hidden in a single field. This is why the threshold reformulation of Section 5 escapes it.

All statements were checked exactly, using rational arithmetic (for all $N \le 120$, and $N \le 300$ when $k=1$) and, for $N \le 7$, by solving the zero-sum search game as a linear program, whose value equalled $h(N,k,n)$ in every case. The code is in `scripts/17_part2_verify.py`.

5 Certifying a threshold assertion (RQ3)

Following [6], define the assorter $a_j = b_j/(2\theta) \in [0, 1/(2\theta)]$. Then $\bar{a} > 1/2 \iff p > \theta$, so the assertion becomes a one-sided test of $H_0 : \bar{a} \le 1/2$ for a bounded finite population. The observation unit is the field: an expert verifies every cell in a sampled field, the field mean of the assorters is observed, and with equal field sizes the population mean over fields equals $\bar{a}$.

Uniform sampling. ALPHA constructs a non-negative supermartingale under H_0 for sampling without replacement [6]. Rejecting when it exceeds $1/\alpha$ controls the false-certification probability at α for any data-dependent stopping time, by Ville's inequality.

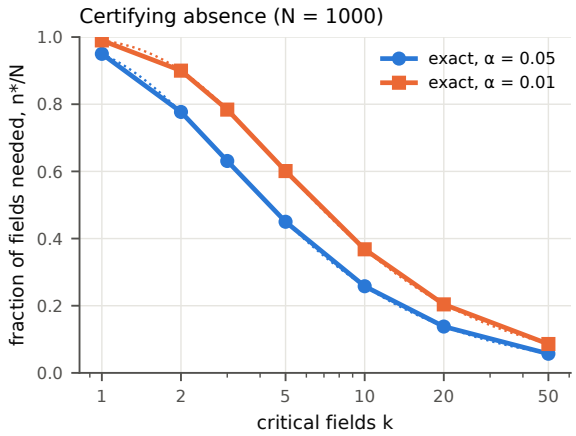


Figure 1: Fraction of fields n^*/N that any strategy without location information must examine to certify the absence of k critical fields ($N = 1000$; exact hypergeometric values, solid; sufficient closed form $1 - \alpha^{1/k}$, dotted). The values depend only weakly on N.

Adaptive sampling. Let q_t be any predictable distribution over the unsampled fields that is strictly positive on all of them. Following Shekhar et al. [7], the importance-weighted observation $Z_t = \pi f(I_t)/q_t(I_t)$ has conditional mean equal to the population mass not yet sampled, $\sum_{i \text{ unsampled}} \pi f(i)$. Under H_0 this is at most $\mu_t = m_0 - \sum_{s<t} \pi f(I_s)$, the mass the null allows to remain. A betting process $W_t = \prod_s (1 + \lambda_s(Z_s - \mu_s))$ with predictable, non-negative bets, bounded so that wealth cannot become negative, is then a test supermartingale under H_0, whatever the classifier does. We use their prop-MS sampler, $q_t \propto S$ with a floor, where S is the field's mean classifier score; approximate-Kelly bets; and, as a variant, a zero-mean control variate built from S. Validity requires only that q_t is known and positive and that labels come from verification. The classifier affects efficiency, not validity.

Designs compared. All designs are run on the same specimen:

- ALPHA with uniform sampling;
- betting with uniform sampling;
- betting with classifier-driven sampling (prop-MS);
- classifier-driven betting with the control variate;
- a deliberately invalid *uncorrected* design, which prioritises fields by classifier score and analyses them with unweighted ALPHA as if they had been sampled uniformly;
- conventional 100- and 200-cell counts, which call "above" when the observed fraction exceeds θ and carry no guarantee.

6 Experimental design

Pre-registration. Hypotheses, decision rules, seeds and analysis scripts were frozen as git tags before the corresponding data were generated. There were four tags: Part I (with one amendment made before any confirmatory data existed), Part III with the external dataset, spatial clustering and a defensive mixture, and real-specimen spatial structure. Exploratory runs used separate seed families and are labelled as exploratory. All deviations from the protocol and from the reimplemented method are logged (Appendix B).

Data.

- *KU-Optofil* [15] (CC BY 4.0). We used 28,280 single-cell images from 275 identified patients (13 classes, 368 px crops), excluding 3,204 cells without patient identifiers. Blasts come from only 11 patients (489 blasts), a limitation we return to below. Cells were assigned to five patient-grouped folds, and every cell's classifier score is cross-fitted, i.e. produced by a model that never saw its patient.
- *RV-PBS* [16]. Real 1000× fields photographed with a phone camera; 752 annotated cells (153 blasts) from 583 images. Its annotation is class-partial, so field-level proportions are not identified. We use it only as an acquisition-shift probe: to measure classifier error, and to build specimens from its labelled crops.
- *BoneMarrowWSI-PediatricLeukemia* [17] (CC BY 4.0). Whole-slide bone-marrow aspirate images, in which 228 regions of interest of about 225 μm were exhaustively annotated: every haematological structure was boxed and consensus-labelled by several experts. We used only the annotations.

Representation and classifier. Cells were embedded with a frozen, publicly available encoder (DINOv2 ViT-S/14 [18], 224 px), which was not fine-tuned on these data. A logistic probe (blast vs. rest) was cross-fitted by patient fold and reached a pooled out-of-fold AUC of 0.979. The thesis this paper derives from does not compare encoders.

Composite specimens (Parts I and III).

- *Part I*: $N = 200$ fields of $m \in \{1, 5, 10, 20\}$ cells, with $k \in \{1, 2, 5, 10, 20\}$ critical fields each containing one blast. A field is represented by the mean of its cells' embeddings (max-pooling and single-cell representations were run as sensitivity analyses). Cells come from the evaluation fold.
- *Part III*: 100 fields of 5 cells ($M = 500$), with the blast proportion fixed exactly. Blasts are drawn without replacement from all 489 blasts, and background cells from blast-free patients. This is allowed because scores are cross-fitted.

InfoDPP-PAC reimplementation. We implemented the paper's selection objective, greedy blended gain and halting rule (Eqs. 3–7 of [4]) with the published tuned weights. The paper reports neither the threshold τ, the constant ρ, the kernel nor the teacher for our domain. We therefore calibrated τ on blast-free specimens from other folds to reproduce the paper's reported operating point (about 16% of the budget), and set $\rho = 1$. Our implementation consequently reproduces the paper's *cost*, but not its nominal PAC guarantee, which we do not claim. Baselines were uniform sampling at the same per-episode cost (paired), a fixed field order (composites have no geometry, so this is not a spatial raster), confidence-based stopping, and exhaustive examination.

Statistics.

- Miss rates: Wilson intervals, exact McNemar tests on paired episodes, and fold-clustered bootstrap intervals.
- False-certification rates: Clopper–Pearson intervals, and the pre-registered rule of rate ≤ 0.05 with a one-sided exact binomial test not rejecting ≤ 0.05.
- Paired cost comparisons: one-sided Wilcoxon signed-rank tests with Holm correction.
- Spatial clustering: an intraclass correlation (ICC) with permutation and patient-bootstrap inference.

The level is $\alpha = 0.05$ throughout.

7 Results

7.1 Information-saturation stopping (RQ1)

The confirmatory run comprised 24,000 episodes: 1,000 per (m, k) cell, and 3,000 for the two cells of the equivalence hypothesis. Figure 2 and Table 1 summarise the results.

The stopping point is content-blind (H1a, supported). The median stop was 32–33 of 200 fields in every cell. Across rarity levels, the median varied by at most one field within each dilution level; the pre-registered tolerance was two fields. This is what the halting statistic implies: it depends on the embeddings and on the selected set, not on any relevance or label term [4].

The halting signal decouples from critical evidence (Kill-test 1, supported). The AUC with which the information-gain term separates critical from non-critical fields fell from 0.778 ($m = 1$) to 0.593, 0.552 and 0.525 ($m = 5, 10, 20$). The teacher relevance score, which the halting rule does not use, stayed at 0.977, 0.943, 0.920 and 0.896.

Premature stops. At $k = 1$, InfoDPP-PAC missed the critical field in 0.315, 0.681, 0.759 and 0.805 of specimens for $m = 1, 5, 10, 20$. Uniform sampling at the same per-episode cost missed it in 0.822, 0.830, 0.846 and 0.828 of specimens, in agreement with $1 - n/N$ (H1d, supported; all $|z| < 1.5$).

- The advantage of saturation-based selection shrinks with dilution for every tested rarity level (H1b, supported). The contrast $\Delta(20, k) - \Delta(1, k)$ had fold-clustered 95% intervals of $[0.39, 0.57]$, $[0.51, 0.59]$, $[0.26, 0.33]$ and $[0.07, 0.17]$ for $k = 1, 2, 5, 10$.
- The pre-registered claim of *no* advantage at $m \geq 10$ was **not supported** (H1c). At $m = 10$ the difference was -0.087 $[-0.107, -0.067]$, outside the ± 0.05 margin; in a secondary analysis it varied across patient folds from -0.03 to -0.16. At $m = 20$ it was -0.023 $[-0.043, -0.004]$, inside the margin.

The correct statement is therefore that the advantage decays with dilution and becomes negligible between 10 and 20 cells per field. Relative to the proposal's hypothesis that the gap would widen with rarity, the governing variable was dilution, not rarity: rarity drove the *absolute* miss rate of every method.

Exploratory analyses. Max-pooled field representations retained somewhat more signal (difference -0.09 at $m = 10$), but still missed 73% of single-critical-field specimens. Varying the operating point from 5% to 50% of the specimen never produced a difference larger than 0.10. Confidence-based stopping was cheaper (median 8–41 fields) and missed more often (0.78–0.88 at $m \geq 10$, $k = 1$).

7.2 Certified threshold decisions (RQ3)

The confirmatory run used 12,000 specimens, with 1,000 per value of p for each threshold. Figure 3 and Table 1 summarise the results.

- **Validity (H3a).** All four valid designs passed at both thresholds. False certification at $p = \theta$ ranged from 0.001 to 0.022.
- **Uncorrected adaptive selection (H3b).** The uncorrected design certified falsely in 0.993 (ICC, $\theta = 0.10$) and 0.889 (WHO, $\theta = 0.20$) of boundary specimens.
- **Cost (H3c, H3d).** Classifier-driven betting needed fewer cells than ALPHA with uniform sampling at every $p > \theta$ (all Holm-adjusted $p < 10^{-59}$). The median cells examined were:

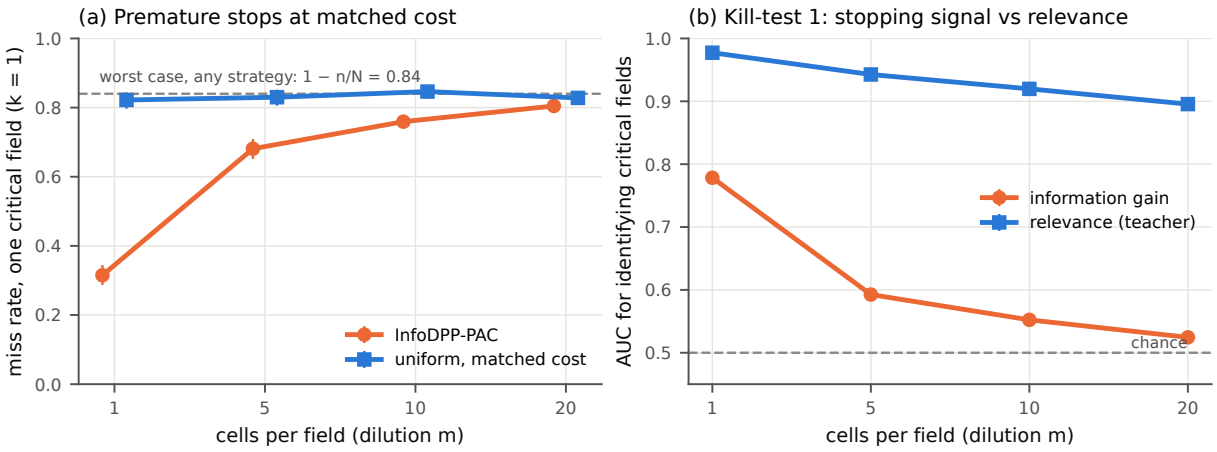


Figure 2: Part I (RQ1), confirmatory run. (a) Miss rate with one critical field, for InfoDPP-PAC and for uniform sampling at the same per-episode cost (95% Wilson intervals). Dashed: the worst-case bound $1 - n/N$ for any strategy without location information (Theorem 1). (b) AUC of the information-gain term that drives stopping, and of the teacher relevance score, for ranking critical above non-critical fields (95% bootstrap intervals).

  - at $\theta = 0.10$: 220 vs. 390 ($p = 0.12$), 75 vs. 165 ($p = 0.15$), 40 vs. 60 ($p = 0.20$);
  - at $\theta = 0.20$: 335 vs. 445 ($p = 0.22$), 145 vs. 250 ($p = 0.25$), 60 vs. 105 ($p = 0.30$).

  Cost fell monotonically with the margin $|p - \theta|$. Near the boundary, any valid design approached a census.

- **Conventional counts.** A 100-cell count called "above" in 0.41 ($\theta = 0.10$) and 0.43 ($\theta = 0.20$) of specimens lying exactly on the threshold. It also called "above" in 0.14 and 0.24 of specimens 0.02 below the threshold.

7.3 Where the assumptions break (RQ4)

Classifier error is not exchangeable (exploratory, KU-Optofil).

- Blast recall at 95% specificity varied across the 11 blast patients from 0.39 to 1.00 (homogeneity $\chi^2_{10} = 61.1$, Monte-Carlo $p = 0.0002$).
- False-positive rates across blast-free patients were overdispersed 6.6-fold relative to a common binomial rate.
- Errors concentrated in the immature granulocyte series and in lymphoid cells.
- Within cell class, errors were associated with stain colour and contrast.

Certification remained valid even when blasts were drawn only from the low-recall patients (false certification ≤ 0.007). Classifier-driven certification at $p = 0.15$ then needed 2.3 times as many cells as with blasts from high-recall patients (140 vs. 60).

Acquisition shift (confirmatory, RV-PBS). Applied unchanged to RV-PBS crops, the KU-trained probe's blast recall fell from 0.906 to 0.340 [0.250, 0.421] (H4a, supported), and its AUC from 0.979 to 0.786. Its false-positive rate did not rise: 0.058 [0.037, 0.081] vs. 0.050 (H4b, **not supported**). The certificate built on these shifted scores remained valid, with 3 false certifications in 1,000 specimens (H4c, supported). Efficiency, however, reversed. On fresh episodes (H6d, supported), classifier-driven sampling needed a median of 195 cells, against 110 for uniform

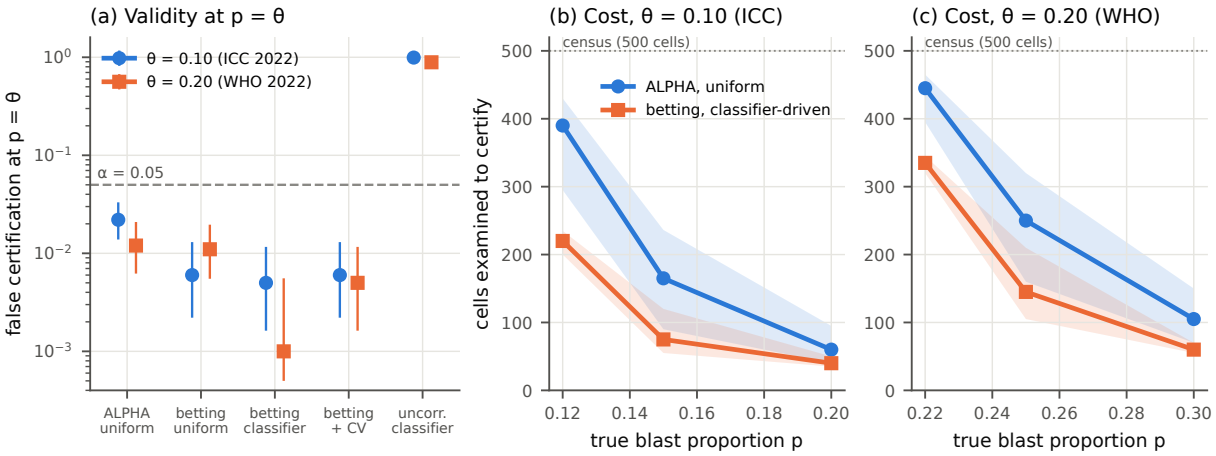


Figure 3: Part III (RQ3), confirmatory run (500-cell specimens; $\alpha = 0.05$; designs paired on the same specimen). (a) False-certification rate at $p = \theta$ with 95% Clopper–Pearson intervals, on a log scale. CV: control variate. The uncorrected design ranks fields by classifier score but analyses them as if uniformly sampled. (b, c) Median cells to certification, with interquartile bands; specimens that never certify are counted as 500 cells.

sampling, at $p = 0.20$ (paired Wilcoxon $p = 1.5 \times 10^{-54}$). A plausible explanation is that a classifier which misses most blasts steers the sample away from them.

A defensive mixture (confirmatory). We mixed classifier-driven sampling with 50% uniform sampling ($q_t = \frac{1}{2}q_t^{\text{prop}} + \frac{1}{2}q_t^{\text{unif}}$). The test remained valid on both sources (false certification 0.001–0.014; H6a). In distribution, the mixture retained 58–73% of the classifier-driven saving (H6c, supported). Under shift it reduced the excess cost at $p = 0.20$ from +77% to +23% (ratio 1.23, 95% CI [1.11, 1.30]). This missed the pre-registered bound of 1.20, so H6b was **not supported**. With a 25% uniform share, the mixture gave essentially no protection under shift (ratio 1.73).

Spatial clustering and the unit of analysis. We placed blasts in a few "hot" fields of 5 cells, at nominal densities of 0.6 and 1.0, so that clustering was strong (E3.1, confirmatory, 15,000 specimens).

- The three field-level designs remained valid in all 36 pre-registered cells, with a maximum false-certification rate of 0.042 (H5a).
- Analysing the cells of uniformly sampled fields *as if each had been drawn individually* was valid when blasts were scattered (0.029 and 0.041). It certified falsely in 0.327 [0.298, 0.357] ($\theta = 0.10$) and 0.478 [0.447, 0.509] ($\theta = 0.20$) of boundary specimens under the strongest clustering (H5b).
- Clustering doubled the cost of uniform field sampling, from 160 to 345 cells at $p = 0.15$, $\theta = 0.10$ (H5c). It lowered the cost of classifier-driven sampling, from 80 to 55 cells, because the classifier finds the hot fields.

Clustering in real smears (confirmatory, BoneMarrowWSI). The eligible data comprised 75 patients, 116 fully labelled regions of interest, 9,375 nucleated cells and 5,841 blasts.

- Blasts were clustered within regions (H7a, supported). The intraclass correlation across the four quadrants of a region (about 113 μm) was 0.032 [0.016, 0.050] (permutation $p = 0.024$). At the 4×4 scale (about 56 μm) it was 0.098 (permutation $p = 0.0005$). Nearest-neighbour blast enrichment was also significant ($p = 0.0005$).

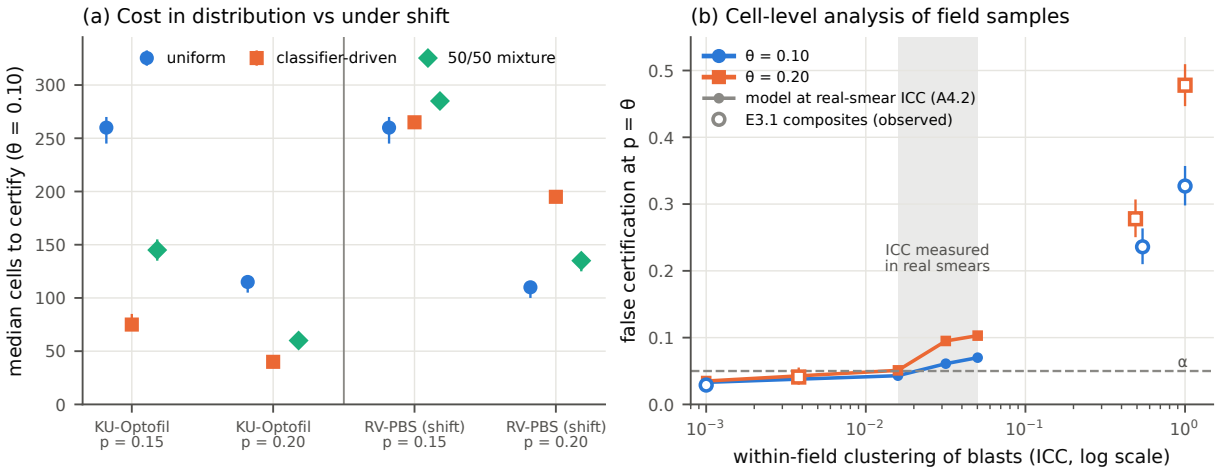


Figure 4: RQ4. (a) Median cells to certify $p > 0.10$ in distribution (KU-Optofil, cross-fitted scores) and under acquisition shift (RV-PBS, KU-trained probe); 95% bootstrap intervals. All designs keep false certification ≤ 0.014. (b) False certification at $p = \theta$ when the cells of sampled fields are analysed as individual draws. Open markers: composites with scattered or hot-field arrangements (5 cells per field). Lines: model at the clustering measured in real bone-marrow regions (17 cells per field). Grey band: 95% CI of the raw ICC, which is biased upward. ICC values of 0 are plotted at 10^{-3}.

- No difference between the two regions of a patient was detected.
- After the run we noticed that the permutation null of the ANOVA estimator is not centred at zero: its mean is 0.018 at the quadrant scale. The excess clustering (about 0.014) is therefore smaller than the raw estimate. The permutation test is unaffected. We report this as an exploratory correction.
- In a pre-specified Beta-binomial model with 17 cells per field, the measured clustering implies that the cell-level mis-analysis certifies falsely in 0.043–0.070 ($\theta = 0.10$) and 0.051–0.103 ($\theta = 0.20$) of boundary specimens across the ICC's confidence range. Field-level analysis stayed ≤ 0.014.
- On the same ICC scale, the E3.1 stress arrangements (0.5–1.0) are one to two orders of magnitude more clustered than these monolayer regions (Figure 4b).

8 Discussion

What is guaranteed, and by what. Our results separate three quantities that are often reported together:

- *Model confidence* about examined fields, which says nothing about unexamined ones.
- *Information saturation*, which concerns the selector's representation of the specimen, not the decision-relevant evidence. Its association with that evidence can disappear, as it did here once cells were pooled into fields.
- *Design-based certification*, which bounds the probability of a false threshold decision for the specific specimen, provided the selection probabilities are known and positive and the labels come from verification.

Only the third is a guarantee about what was not examined, and it held in every condition we tested, including a classifier that lost two thirds of its blast recall. Classifier quality determined only how quickly certification was reached, and could make it slower than uniform sampling.

Table 1: Pre-registered hypotheses and verdicts. All confirmatory data were generated after the corresponding protocol was frozen.

RQ	ID	Hypothesis (abridged)	Verdict
1	H1a	InfoDPP-PAC stop point independent of rarity	supported
1	H1b	Advantage over matched-cost uniform shrinks with dilution	supported (4/4)
1	H1c	No advantage (± 0.05) at $m \geq 10$, $k = 1$	**not supported**
2	H1d	Uniform miss rate equals $1 - n/N$	supported
1	KT1	Information gain and relevance are separable signals	supported
3	H3a	All four designs valid at both thresholds	supported
3	H3b	Uncorrected adaptive selection invalid	supported
3	H3c	Classifier-driven sampling cheaper than uniform	supported
3	H3d	Cost decreases with the margin	supported
4	H4a	Blast recall drops under acquisition shift	supported
4	H4b	False-positive rate rises under shift	**not supported**
4	H4c	Certificate valid with shifted side information	supported
4	H5a	Field-level tests valid under clustering	supported
4	H5b	Cell-level analysis of field samples invalid under clustering	supported
4	H5c	Clustering raises the cost of valid certification	supported
4	H6a	Defensive mixture valid	supported
4	H6b	Mixture caps loss under shift below 20%	**not supported**
4	H6c	Mixture keeps $\geq 50\%$ of in-distribution saving	supported
4	H6d	Cost reversal under shift replicates	supported
4	H7a	Blasts spatially clustered in real smears	supported

Practical implications.

- Certify a proportion against a stated threshold rather than the presence or absence of rare cells; the latter requires a near-census (Theorem 1).
- Record selection probabilities, and never analyse adaptively selected fields as if they had been sampled uniformly.
- Analyse at the unit that was sampled. Treating the cells of a sampled field as independent draws is anti-conservative whenever blasts cluster. The effect is modest in the evenly dispersed monolayer we measured and severe under strong clustering.
- Mixing a fixed uniform share into classifier-driven sampling limits, but does not remove, the efficiency loss under shift.

Limitations.

- Most specimens are composites of single-cell images from one laboratory. A composite specimen mixes cells from several patients, so it is not a real patient's smear, and all blasts in Parts I and III come from 11 patients. Results varied across patient folds.
- The external blood dataset is small, lacks patient identifiers, and was cropped by our own rule, which confounds stain and camera effects with framing.
- The real-smear spatial analysis uses pediatric bone-marrow aspirates, not peripheral blood. Its regions were chosen in the evenly dispersed monolayer, so clustering elsewhere in a smear may be larger. The consequence for validity at the measured clustering comes from a model, not from observed certification rates.
- Verification is simulated with expert labels ($\beta = 0$).

- InfoDPP-PAC's unreported hyperparameters were calibrated to its reported cost, so our reimplementation may differ from the authors' configuration.
- We used a single frozen encoder.
- We did not implement prediction-powered e-values [14].

9 Conclusion

When an adaptive system decides how much of a blood smear to examine, information saturation does not certify what remains unexamined. Without location information, certifying the absence of rare evidence requires examining nearly all of the specimen, whatever the strategy. Threshold decisions about the blast proportion, by contrast, can be certified with finite-sample, anytime-valid guarantees whose validity rests on the sampling design and the unit of analysis rather than on the classifier. Extending this analysis to fully annotated peripheral-blood smears, and to verification by haematologists, is the most important next step.

Data and code availability. All code, the pre-registration and its git history, and the analysis outputs are available at [repository URL]. KU-Optofil, RV-PBS and BoneMarrowWSI-PediatricLeukemia are public datasets used under their respective licences.

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A Proof of Theorem 1

Fix x and a realisation u of the examiner's independent randomness U.

Step 1 (the all-negative path). Consider the run in which every examined field is negative. Every input to the policy is then determined by (x, u), so the run examines a fixed sequence $a_1(u), \ldots, a_{n'}(u)$ with $n' \le n$. Let $A(u)$ be the set of these fields.

Step 2 (the miss event is determined by $A(u)$). If $C \cap A(u) = \emptyset$, every field on the all-negative path is negative under C, so the actual run follows that path and misses C. Otherwise, let t be the first index with $a_t(u) \in C$. Up to step $t-1$ all observations are negative, so the actual run follows the all-negative path through step t, examines $a_t(u) \in C$, and does not miss. Hence $\mathbb{P}_\pi(M_n \mid x, C) = \mathbb{P}_U(C \cap A(U) = \emptyset)$.

Step 3 (averaging over placements). Let C be uniform over the k-subsets, independent of U; Assumption 1 makes every such C admissible. For fixed u, the fraction of k-subsets disjoint from $A(u)$ is $\binom{N-|A(u)|}{k}/\binom{N}{k} \geq \binom{N-n}{k}/\binom{N}{k} = h(N,k,n)$, using $\binom{N-n}{k}/\binom{N}{k} = \binom{N-k}{n}/\binom{N}{n}$. Averaging over U gives $\mathbb{E}_C \mathbb{P}_\pi(M_n \mid x, C) \geq h(N,k,n)$.

Step 4 (worst case is at least the average). $\max_C \mathbb{P}_\pi(M_n \mid x, C) \geq \mathbb{E}_C \mathbb{P}_\pi(M_n \mid x, C) \geq h(N,k,n)$. Uniform sampling achieves h for every C by Proposition 1.

For a deterministic policy, A is a single set with $|A| \leq n \leq N - k$. Placing C inside $[N] \setminus A$ gives miss probability 1. □

B Deviations from the protocol and from the reimplemented method

- **InfoDPP-PAC.** We used DINOv2 ViT-S/14 instead of the paper's Phikon (H&E) encoder, because there is no established open Wright–Giemsa encoder. τ was calibrated to the reported operating point, and $\rho = 1$. A field is the mean of its cells' embeddings. The "fixed-order" baseline is not a spatial raster.
- **Part III.** An early exploratory construction fed importance-weighted draws into ALPHA's unweighted null-mean update, which is not a valid supermartingale. It was withdrawn before the confirmatory protocol and replaced by the construction of Shekhar et al. [7].
- **Part I analysis.** An exploratory batch-overwrite bug, found before the protocol was frozen, had caused early estimates to use three of five folds. It was corrected.
- **Real-smear spatial analysis.** The analyses were run with byte-identical frozen scripts, in a different compute environment because of time limits (SHA-256 hashes recorded). The bias of the ICC estimator was noticed after the run and is reported as exploratory.
- **Not implemented.** Prediction-powered e-values; haematologist verification.